Integration Hub Implementation

Hybrid deployment with CloudHub 2.0 + Runtime Fabric on OpenShift
Secure connectivity to on-prem systems • Modernization from a legacy ETL stack

Built on

Runtime Fabric

on Red Hat OpenShift

Prepared by: <Your Name> • <Role/Team>
For: <Client/Stakeholders> • <Date>

OVERVIEW

What was delivered

A production-grade Integration Hub that standardizes integrations, enables API reuse, and supports workloads across both public cloud and a client-controlled private cloud environment.

- Hybrid runtime: CloudHub 2.0 for elastic cloud workloads; Runtime Fabric on OpenShift for regulated / low-latency workloads
- Secure, private connectivity to on-prem systems using private networking patterns (IPsec VPN) and API security controls
- Modernized legacy integrations into an API-led model with shared assets and governance

Faster delivery

- Reusable APIs & templates
- Standard CI/CD patterns
- Clear deployment decision rules

Stronger security

- Private network extension
- Policy-based API security
- Centralized secrets & logging

Operational resilience

- Active monitoring & alerts
- Horizontal scaling
- Controlled change + rollback

CONTEXT

Common integration pain points addressed

Point-to-point complexity

- Hard to change safely
- Duplicate transformations
- Inconsistent security

Solved via standard APIs + shared runtime patterns

Batch-first legacy patterns

- Limited real-time capability
- Long feedback cycles
- Tightly-coupled schedules

Solved via standard APIs + shared runtime patterns

Operational fragmentation

- Multiple runtimes & toolchains
- Limited visibility
- Slow incident response

Solved via standard APIs + shared runtime patterns

SOLUTION

Integration Hub = reusable APIs + shared platform services

A consistent “front door” for integrations: discoverable APIs, governed standards, and repeatable deployment patterns across cloud and private environments.

Experience APIs | Process APIs | System APIs | Core Systems

API design & reuse

- System / Process / Experience layering
- Asset publishing & versioning
- Contract-first specs

Connectivity & integration

- Enterprise connectors
- Transformation + orchestration
- Event and batch patterns

Security & governance

- OAuth2/JWT enforcement
- Rate limiting & SLAs
- Standards via API Governance

Operations & observability

- Centralized monitoring
- Log search & alerts
- Runbooks + SLOs

ARCHITECTURE

One control plane • Two runtimes • Secure connections

Anypoint Platform (control plane)

API Manager • Exchange • Runtime Manager • Monitoring

CloudHub 2.0 (Private Space)

Managed iPaaS runtime for elastic cloud workloads

- API gateway / policies
- Experience APIs
- Process APIs
- Schedulers / events

Ingress via LB + TLS (custom certs)

Runtime Fabric (RTF) on OpenShift

Client-controlled Kubernetes runtime in private cloud DC

- System APIs
- Internal-only services
- API proxy / ingress
- High-throughput jobs

Ingress via OpenShift Routes / Ingress

IPsec VPN

Private link

Client on-prem systems (apps, databases, message brokers) behind corporate firewall

HYBRID

Choosing the right runtime per workload

A simple, explainable rule-set keeps teams aligned and reduces design churn.

CloudHub 2.0

Best fit

- Externally-facing APIs needing managed ingress + elastic scale
- Integrations with primarily cloud endpoints
- Workloads where managed operations are preferred
- Teams optimizing for speed to delivery

Watch-outs

- Data residency constraints
- Extreme low latency to on-prem
- Strict client-controlled runtime requirements

Runtime Fabric on OpenShift

Best fit

- Workloads requiring client-managed infrastructure or bespoke networking
- High-throughput or low-latency integrations near on-prem systems
- Data sovereignty and regulated processing
- Integrations needing custom ingress, routing, or cluster controls

Watch-outs

- Requires cluster operations (OpenShift/RTF)
- Capacity management and patching planning
- Ingress and cert management design

SECURITY

Secure connectivity to on-prem systems

Network security

- Private Space network extension for CloudHub 2.0 workloads
- Site-to-site IPsec VPN option for private connectivity
- Segmentation: inbound/outbound firewall rules per integration zone
- No direct public exposure for internal-only APIs

Transport security

- TLS everywhere (custom certs)
- mTLS where required (service-to-service)
- Certificate rotation runbooks

API security controls

- OAuth2 / JWT token validation policies applied consistently
- Rate limiting & SLA tiers for consumer management
- Central policy management via API Manager
- Audit-ready traces via monitoring + log retention

Secrets & identity

- Environment-specific configs
- Least-privilege access to platform
- Rotation and access reviews

MIGRATION

A phased migration that de-risks cutover

Designed to preserve business continuity while modernizing capabilities.

1. Discover

Inventory flows
Classify criticality
Identify quick wins

2. Design

Define API contracts
Layered model
Choose runtime per flow

3. Build

Reusable connectors
Shared templates
Automated tests

4. Migrate

Parallel run
Strangler cutover
Data validation

5. Decommission

Retire old jobs
Remove licenses
Harden runbooks

Risk controls baked into delivery

- Contract tests to protect consumers
- Parallel run + reconciliation reports
- Blue/green or rolling deploys with rollback
- Decommission checklist and audit trail

DELIVERABLES

Integration Hub deliverables (examples)

Replace the examples below with your actual assets and names.

Standard API templates

- Maven archetype / starter repo
- Logging + correlation IDs
- Error handling framework

Observability baseline

- Dashboards per API & runtime
- Alert thresholds & runbooks
- Log search patterns for incidents

Shared connectors & accelerators

- Reusable connectors to core apps
- Canonical data model transforms
- Batch-to-event modernization patterns

Security baseline

- OAuth2/JWT policy bundles
- Rate limiting defaults
- TLS/mTLS certificate procedures

Platform guardrails

- Naming + versioning standards
- CI/CD policy checks
- Deployment decision matrix

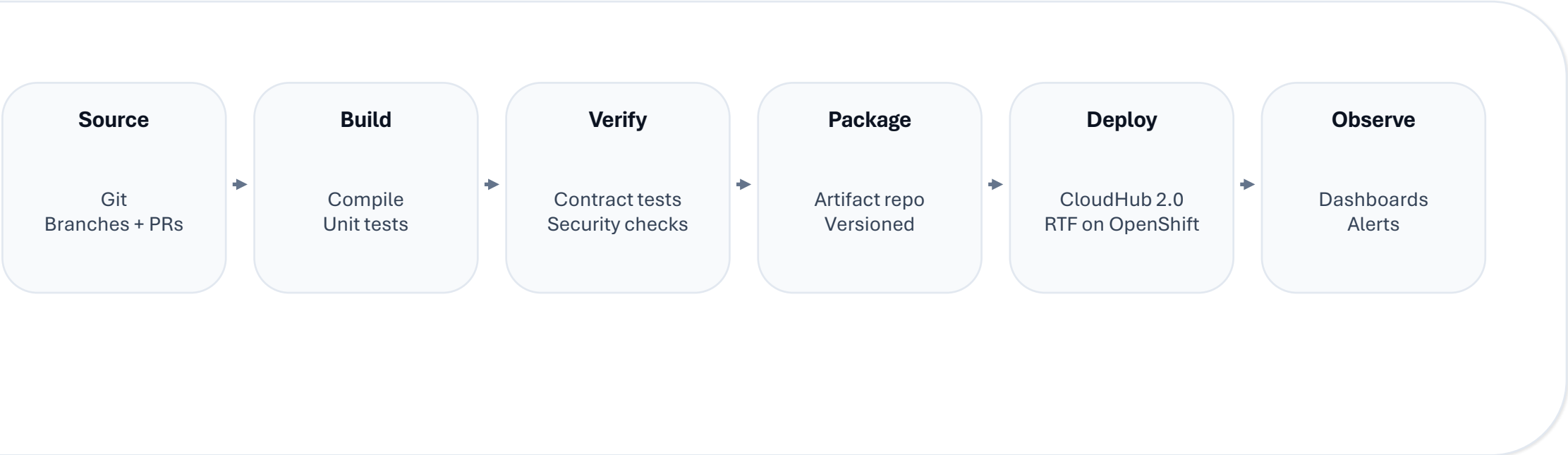
Operations enablement

- On-call playbooks
- Change management steps
- Backup/restore & DR drills

DEVOPS

A single delivery pipeline for both runtimes

Same controls, same quality gates — deploys route to CloudHub 2.0 or RTF based on configuration.



Key practices: configuration-by-environment • automated policy application • standardized logging/correlation • repeatable rollback

OPERATE

Operating model that scales across teams

Platform team responsibilities

- Define standards (naming, versioning, security policies)
- Operate shared services (connectivity, monitoring, CI/CD templates)
- Own runtime foundations (private spaces, RTF clusters, base images)
- Enablement: patterns, reviews, training, office hours

Domain/product team responsibilities

- Build and own APIs in their domain
- Meet standards & pass automated checks
- Run and support their integrations

Governance loop

Design	Specs + standards checks
Publish	Exchange assets Versioned
Enforce	API policies SLA tiers
Monitor	Dashboards Alerts
Improve	Post-incident Backlog

VALUE

Quantify outcomes (replace placeholders)

Use the metrics your stakeholders care about. Examples below are placeholders — insert real numbers.

Integration delivery lead time

↓ __ %

Reusable API adoption

__ APIs reused

Incidents / MTTR

↓ __ %

Outcome story

- Teams deliver new integrations faster because common patterns are already built and governed.
- Operational resilience improves with standardized monitoring, alerts, and runbooks across both runtimes.
- Hybrid placement optimizes cost, latency, and compliance by putting each workload in the right environment.

ROADMAP

Roadmap (example)

Edit these items to match your client and your delivered milestones.

Now (0–30 days)

- Finalize standards and templates
- Baseline monitoring + dashboards
- Prioritize next migration wave

Next (30–90 days)

- Migrate high-value flows
- Expand reusable System APIs
- Harden HA/DR and runbooks

Later (90–180 days)

- Optimize cost and scaling
- Introduce event-driven patterns where needed
- Decommission remaining legacy jobs

APPENDIX

Glossary (for non-technical audiences)

Integration Hub	A standardized platform for building, deploying, and operating APIs/integrations with shared patterns.
CloudHub 2.0	Managed iPaaS runtime for deploying Mule apps as lightweight containers in the cloud.
Private Space	A logically isolated CloudHub 2.0 space that can be connected to a private intranet for a single private network.
Runtime Fabric (RTF)	Customer-managed runtime that deploys Mule apps and API proxies to a Kubernetes cluster.
OpenShift	A Kubernetes platform used here to host Runtime Fabric in a client-controlled environment.
IPsec VPN	A standard method for creating encrypted site-to-site network connectivity.